

WONDIMAGEGN TESFAYE BESHAH

Geospatial Developer

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SUMMARY

With over 14 years of experience, I have successfully delivered 28 geospatial development projects across 13 countries for esteemed clients, including NASA, UN-Habitat, and Fortune 500 companies such as Comcast and Duke Energy. My deep expertise, detail-oriented approach, and passion for geospatial application development have enabled me to craft innovative solutions that drive impactful results on a global scale.

PROFESSIONAL EXPERIENCE

Senior Geospatial Developer

Rina North America (Patrick Engineering Inc.), Remote | 25 Aug 2021 – 17 Jan 2025

- Delivered code focused on scalability, testability, supportability, and maintainability.
- Contributed to application development for Comcast, Duke Energy, Spectrum, and ADP clients.
- Developed web applications using Esri ArcGIS SDK for JavaScript, ArcGIS Enterprise, Flask, AWS, Angular, React, TypeScript, Experience Builder, Web App Builder, D3, ArcSDE, MSSQL and Oracle SQL Developer.

Research Assistant

Mississippi State University, MS | 15 Jun 2018 – 25 Jul 2021

- Developed remote sensing algorithms for water quality analysis in the Gulf of Mexico.
- Conducted time series analysis, UAV and satellite image processing.
- Built geospatial visualization tools and dashboards using NodeJS, Mapbox, Vector Tiles, GDAL, Pandas, ExtJS, Web Sockets, OpenLayers, Python, JavaScript, and R.

GIS Developer (Consultant)

International Center for Tropical Agriculture (CIAT), Nairobi, Kenya | 29 Aug 2017 – 25 May 2018

- Developed a web-based agricultural land suitability, zonal statistics, and land similarity tool.
- Optimized ArcGIS ArcPy desktop scripts for speed and efficiency.
- Built custom geoprocessing services and widgets for ArcGIS Web App Builder.

GIS Developer (Consultant)

RCMRD – European Union Project, Nairobi, Kenya | 05 Jun 2017 – 01 Jan 2018

- Developed an open-source web-based mapping application to visualize, compare, analyze, and download forest-related data in Kenya, Tanzania, Uganda, and Mozambique.

Python Developer (Consultant)

UN-Habitat, Nairobi, Kenya | 09 Jun 2016 – 25 May 2018

- Supported Social Tenure Domain Model (STDM) initiatives through system development and training.
- Enhanced existing STDM modules and data and configuration migration tools.
- Designed STDM releases based on country implementation experiences and emerging trends.

GIS Programmer (Consultant)

UN Habitat, Nairobi, Kenya | 15 Feb 2016 – 08 Jun 2016

- Created a Software Requirement Specification Document, UML Diagram, and Database Diagram for the Turkana County Land Information Management System.
- Customized STDM by modifying Python-based modules.

- Developed Parcel Details Module and integrated it into STDM.
- Authored user manuals and trained county staff on system usage.

GIS System Developer

RCMRD – NASA SERVIR Project, Nairobi, Kenya | 14 Jul 2014 – 16 Feb 2016

- Developed web mapping applications for the SERVIR Eastern Africa project led by NASA SERVIR Global.
- Conducted training on web mapping.

General Manager & Website Administrator

Sewite PLC, Ethiopia | 01 Aug 2013 – 31 Mar 2014

- Designed and managed websites such as ethiostay.com, ethioinvest.com, and chinaafricatrade.com.
- Enhanced website performance, SEO ranking, and user engagement.
- Provided strategic business recommendations and oversaw financial forecasts, HR, and PR activities.

Cultural Exchange Coordinator & Website Administrator

Bright Future School, Ethiopia | 01 Jul 2011 – 30 Jul 2013

- Coordinated cultural exchange programs and guided students through the exchange process.
- Designed projects, wrote proposals, and conducted fundraising.
- Developed and managed the school's website (www.brightfutureschool.com).

EDUCATION

Ph.D. in Earth and Atmospheric Sciences (Ongoing) – Mississippi State University, MS (Jun 2018 – Dec 2025)

❖ **Graduate Research Award Winner (Apr 2021)**

MA in Development Studies – Tata Institute of Social Sciences, India (Jun 2008 – May 2010)

BA in Community Development & Leadership – Ethiopia Adventist College, Ethiopia (Oct 2006 – Feb 2011)

❖ **Gold Medal Winner for Academic Performance**

BEd in Geography – Haramaya University (Formerly Alemaya University), Ethiopia (Oct 2000 – Jul 2004)

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, R, PHP, HTML/CSS, TypeScript
- **Web Tools:** Flask, Django, Angular, React, ExtJS, Node.js, Apache Web Server, Apache Tomcat
- **GIS Tools:** ArcPy, GeoServer, PyQGIS, ArcGIS Server, ArcGIS Enterprise, ArcGIS SDK for JavaScript, FME, OpenLayers, Leaflet, Mapbox, Web AppBuilder, Experience Builder
- **Data Analysis:** Pandas, GDAL, PostgreSQL/PostGIS, MSSQL, MySQL, Oracle SQL Developer, NumPy, Scikit-learn, SciPy, OpenCV
- **DevOps:** Git, GitHub, BitBucket, Splunk, Jenkins, Jira
- **Cloud:** Amazon Web Services, Microsoft Azure

CERTIFICATIONS & TRAININGS

- **Cornell Summer Satellite Remote Sensing** – Cornell University, USA (Jun 2019)
- **Object-Oriented Programming in PHP** – Institute of Software Technologies, Kenya (Jun 2015)
- **Python Programming** – Institute of Software Technologies, Kenya (Jun 2015)
- **Geo-Spatial Database Development** – RCMRD, Kenya (Dec 2014)